The Robockey Cup

A Look at Mechatronics Education in 2009



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BY JONATHAN FIENE

Each fall, a group of intrepid students converge in a classroom on the third floor of the University of Pennsylvania's Towne Engineering Building for what they know will be one of the most challenging courses in their academic career. That first class begins with a discussion of where it will end. In particular, we talk about what they will accomplish in the final project, where they will design, fabricate, assemble, program, and debug small teams of autonomous hockey-playing robots, complete with wireless communications, infrared puck sensing, and enough onboard computational power to handle just about any task that can dream up. Most of the students look incredulous; after all, many of them have never built a circuit or written C code before.

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The Rules of the Game

We begin the semester by introducing the final project, and we shall do the same here.

The Rink

The Robockey rink resembles a generously rounded rectangle of approximately 240 × 120 cm, with 50-cm wide goals at each end. The rink is surrounded by 4-cm tall clear polycarbonate walls, and the surface of the rink is painted white with a wax finish to minimize friction.

The Puck

The object of each robot's affection is a 7.6-cm diameter custom-machined acrylic puck (as can be seen in Figures 1 and 2). The puck has three integrated ball casters, which allow it to easily glide the length of the rink. Inside the puck is a pair of lithium-polymer batteries, a small circuit board, and eight 940-nm infrared light-emitting diodes (LEDs) radiating outward. Sensing these LEDs will be the only way that the robots will be able to locate the puck on the field.

The Robots

Each team may field a team of up to three robots that must: 1) be shorter than 13 cm and remain within a 15-cm cylinder at all times; 2) be fully autonomous; 3) carry its own power source; 4) not constrain the puck's motion; 5) not intentionally damage the rink, the puck, or other robots; 6) not maliciously interfere with the wireless communication system; and 7) not emit or intentionally reflect infrared light.

Robot Localization System

A video camera located above the rink is used in combination with ARToolKit [1] to provide position data for each of the robots. To make them visible to the tracking system, each robot is required to have a threaded rod at its uppermost point to which a 12-cm-wide tracking fiducial can be attached (one of these can be seen atop the robot in Figure 1). When a robot is found by the tracking system, the corresponding X and Y location relative to the center of the rink is broadcast over the wireless system.

Wireless Communications

A local wireless network allows the tracking system and game controller to communicate with each robot while also allowing intrateam communication. Each robot in the tournament is assigned a unique 5-B address and must demonstrate an ability to follow single-byte hexadecimal game commands, such as 0xA1 for play, 0xA4 for pause, or 0xA5 for detangle (move randomly).

Game Play

A regulation Robockey match consists of two 2-min periods. Each period begins with a polo-style start, where the puck is placed in the center of the rink, while robots must be behind the lines located 80-cm from the center. Once the play command is issued, the robots are free to move about the field, with the objective of placing the puck into the opposing team's goal. If a goal is scored, the teams return to the starting positions, and the puck is relocated to the center of the rink. If the score is tied after two periods, a third period will commence. If the game is still tied after three periods, a sudden-death shootout is used to determine a winner.

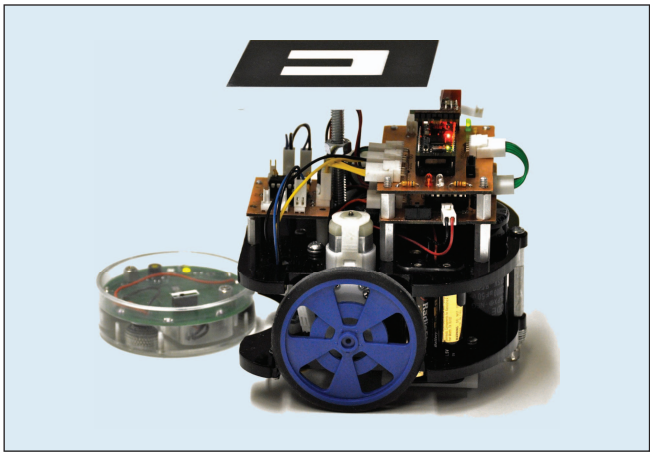


Figure 1. *A student-built autonomous hockey-playing robot.*

Motivation and Pedagogy

Each student brings his or her own set of motivations to the class, yet the added incentive of public competition causes a large percentage of the students to engage the material with significantly more vigor than might have been found otherwise. With the end goal for the semester firmly established, the students are asked to quickly generate a list of topics in which they think they will need to gain competency to succeed at Robockey. This list can seem daunting to many of the students, and it often includes motors, batteries, microcontrollers, programming, wireless communications, sensors, actuators, circuits and systems integration.

To take stock of the current knowledge of this diverse class, which typically draws an equal mix of graduate students and



Figure 2. *Two Robockey robots approaching the infrared-emitting puck.*

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undergraduates from a variety of engineering disciplines, an ungraded concept inventory is given out at the close of the first lecture. In addition to helping the teaching staff to know where students will need the most help, this exercise helps the students to know where they will need to focus their efforts. It is often the case where a few students are already well versed in one or two of the primary mechatronics categories; however, the majority of the students require substantial training in all areas, which include:

1) *Mechanical Design*: mobile–robot kinematics, motors, power transmission, gears, fasteners, basic structures, prototyping, sensors, mechanisms, and solenoids
2) *Electronics*: passive electrical components, voltage, current, resistance–capacitance (*RC*) filters, LEDs, phototransistors, op amps, comparators, semiconductors, transistors, MOSFETs, buttons, switches, inductive loads, digital logic, integrated circuits, noise, signal conditioning, wireless networks, analog-to-digital (A/D) conversion, sensors, and printed circuit board (PCB) design
3) *Programming*: the C programming language, binary, hexadecimal, memory, addressing, input/output, clocks/timers, pulsewidth modulation, A/D conversion, serial communications, interrupts, and event-driving programming.

Although there is an inherent challenge to both learning and teaching such a variety of material within a single semester, there are a number of interlocking elements (sensors, actuators, etc.) that bridge the gaps between the disciplines and help tie the material together. Approximately half of these topics can be taught effectively through traditional lectures and written assignments, whereas the remainder benefits significantly from hands-on experience, where the students can achieve a personal connection to the material [2]. To this end, a series of six small projects are introduced throughout the semester, ranging from a primer on electronic prototyping to a team project to build a small self-balancing robot.

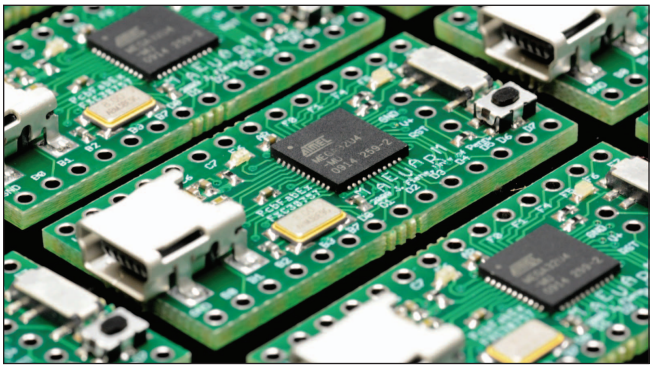


Figure 3. *The M1 custom microcontroller solution.*

Resources

To support this aggressive project and the breadth of course content requires significant infrastructure, including a relatively large dedicated laboratory space, complete with electronic test equipment (multimeters, oscilloscopes, power supplies, and function generators) an assortment of standard electronic components (resistors, capacitors, switches, ICs, etc.), mechanical prototyping and fabrication capabilities (laser cutters, mills, lathes, hand tools, etc.), project storage space, and a dedicated core of teaching assistants, most of whom are alumni from previous semesters.

In addition, one of the most important decisions for a course of this nature is the selection of a suitable microcontroller that the students will be able to embed into their designs. Given experience with a number of different commercially available packages, the decision was made to develop a custom solution based on the Atmel AVR processor. The result of this effort, known as the M1 [3], is shown in Figure 3. This two-layer board measures 1.8 × 4.0 cm and is built around the ATmega32U4 processor, which has 32 K of programmable flash, 1 K of electronically eraseable programmable read-only memory, 2.5 K of static random-access memory, 25 diode-protected general purpose input/output lines, four standard timer/counters, one high-speed timer, 12 channels of 10-b ADC, and a variety of serial communication channels, including universal serial bus (USB) (which can be used for direct PC-to-chip programming, as well as debugging). The chip uses a modified Harvard architecture, takes a nominal supply voltage of 5 V, and uses an 8-MHz external oscillator. The board has a built-in reset button, along with a bootloader/run switch and two LEDs that can be independently controlled. Completely assembled in moderate quantities for less than US$15 per board, this platform is well suited to students who are learning the art of microcontrollers.

Course Components

Having set the pedagogical requirements for the final project, and with an understanding of the existing infrastructure, it is a relatively straightforward process to track backwards through the semester, placing smaller projects, assignments, and lecture topics in such a way that the material roughly fits the hierarchy necessary to build the requisite knowledge base [4]. Looking specifically at the Fall semester of 2009, this process resulted in the following list of laboratories (L) and assignments (A):

- Week 1: Basic electronics
  - L1: Introduction and passive filters—familiarize students with the electronic test equipment in the mechatronics laboratory and then get them working with passive *RC* low-pass and high-pass filters
  - A1: Passive circuit analysis—solve problems related to equivalent resistance, voltage division, *RC* filter time constants, impedance matching, and filter design
- Weeks 2–3: Optoelectronics, sensors, actuators, semiconductors, inductive loads
  - L2: Beacon tracking—provide students with some fundamental experience prototyping simple circuits, using LEDs and phototransistors, and driving dc motors. The final portion of this laboratory requires students to design and built a two-device system, one of which is

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capable of tracking the other using infrared light. Each object must obey certain size constraints, and no physical connection can be made between the two objects. Students are required to demonstrate their system to a member of the teaching staff within a standardized testing environment.
  - A2: Motors and switches—solve problems related to H-bridge switches, dc motors, bidirectional motor control, motor selection, and common motor drive circuits.
- Weeks 3–4: Digital logic, mechanical prototyping, transmissions, gears, feedback control, and op amps
  - L3: Maze—design, construct, and test the drive mechanics and circuitry for a small mobile robot to carry a 500-g mass through a maze using a remote control [see Figure 4(a) for one of the many solutions]. Those enrolled in the graduate section of the course are also required to design and build the controllers to drive the robots through the maze, including the development of a standard communications protocol.
  - A3: Op amps—solve problems including addition and subtraction circuits, integration and differentiation, and transresistance using operation amplifiers.
- Weeks 5–6: Microcontrollers, C, input/output, clocks, timers, A/D conversion
  - L4: Stroboscope—to familiarize students with the M1 microcontroller, they are given the task of building battery-powered strobe lights, where the frequency of the strobe is controlled using a potentiometer and the duty cycle is controlled using a small bank of switches.
  - A4: Localization—exploring one of the more challenging aspects of the Robockey system, students are given the task of developing four different methods by which they could track the location of a robot on a field. Solutions varied from overhead camera-based tracking and indoor GPS to embedding a grid of thousands of LEDs into the field, all flashing at different frequencies.
- Weeks 7–8: USB communication, advanced sensors, accelerometers, interrupts, digital filters
  - L5: Balancing robots—in this first team project, groups of three students design and build a fully contained two-wheeled self-balancing robot using the M1 microcontroller. Students are given a choice of sensors, including microelectromechanical systems-based Freescale MMA7361L triaxial accelerometers. One of the more successful solutions can be seen in Figure 4(b).
- Week 9: Event-driven programming, wireless communication
  - L6: Wireless Morse—to gain an understanding of basic wireless communication between multiple M1 boards, students are tasked with creating a simple wireless Morse code system to send and receive binary information between team members.
- Week 10: PCB design
  - A5: Puck challenge—to familiarize students with the PCB design process, the design of the circuit for the Robockey puck becomes a class project, where the best design will end up on the field. The specifications include geometric constraints, infrared emission characteristics, and power source selection.
- Weeks 11–14: Robockey—the final weeks of the class are devoted to milestones and deliverables for Robockey, including:
  - test of the robot's ability to follow game-play commands
  - demonstration of the robot interpreting localization signals to navigate itself to the center of the rink
  - friendly matches (minimum two per team)
  - preliminary rounds of the double-elimination tournament
  - public festival, including final rounds of the tournament.

The object of each robot's affection is a 7.6-cm diameter custom-machined acrylic puck.

2009 Robockey Results

Mechanical Design

The majority of the groups split their team of robots into two fast-moving "forwards" and a methodical, special-purpose "goalie." Although most teams quickly settled into the relatively standard differential-steering platform (as seen in Figure 1), there were some noticeable exceptions, such as the omnidirectional creation shown in Figure 2. To locate the puck, all of the teams incorporated outward-facing infrared phototransistors near ground level. Many teams relied on mechanical switches to determine when their robot had control of the puck,

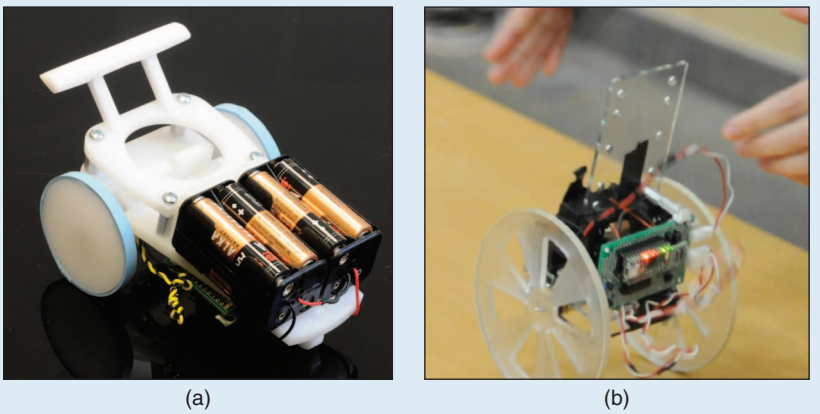


Figure 4. *(a) One of the many maze-going robots built for L3 and (b) one of the balancing robots built for L5.*

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Figure 5. *The 2009 MEAM 410/510 class at the conclusion of the Robockey Cup Tournament.*

while some built small break-beam sensors across the puck concavity. A few of the teams included a kicking mechanism using a dc solenoid and return spring. All teams made liberal use of laser-cut acrylic and mechanical fasteners in the construction of their robots, allowing for rapid iteration, assembly, and disassembly.

Electronics

All of the robots included an M1 microcontroller and a wireless module. Beyond that, most teams chose to use H-bridge ICs to drive the wheel motors, and the remainder of the circuitry focused on power conditioning, analog signal filtering, and digital control. Although a few of the teams chose to design PCBs, most relied on hand-soldering.

Programming

While it was the least visible of the elements in the Robockey robots, the code that students wrote to control their robots was certainly the most decisive factor in the overall performance of the teams. While certain functionality was explicitly required (such as obedience to game play commands), the exact implementation of the control algorithms varied widely from team to team. The programming was often the most difficult of the components for students to grasp and master, and while it was a goal for most, only a handful of the teams were able to successfully implement interrobot strategy (field location, passing, etc.).

Strategy

With the exception of specialized goalies, most of the robots attempted to execute a similar strategy: wait for the play command; search for and drive toward the puck; get the puck, drive toward the opposite goal; and shoot the puck into the goal. With as many as six robots on the field all executing a similar strategy, it was never quite that easy.

Observations and Future Directions

"The student projects went on until 10:00 p.m. The excitement in [the auditorium] was incredible. You could hear the cheers as goals were scored or missed from outside the building. The robot designs were clever and the control algorithms were ingenious."

"It is the capstone course to take for any engineer, because it involves so many interdisciplinary skills from mechanical engineering, electrical engineering, computer science, and robotics. I have been challenged by this course and broadened my breadth of knowledge of all these disciplines."

"I wish more engineering courses were like this, with a mix of theory and a great hands-on component."

Although it can be difficult to quantify the overall effect of any course, the aforementioned quotes speak volumes about the impact that can be achieved.

This semester marked the second running of the Robockey tournament, and given the positive feedback and outcomes, it will likely continue for some time. Moving forward, it is critically important that we stay abreast of current technology, and efforts will certainly be made to reduce the workload, as many students do report that it is one of the most time-consuming courses that they have ever taken.

Additional details for all of the projects and assignments, along with course lecture materials and information on the custom M1 microcontroller board can be found at http://medesign.seas.upenn.edu.

Keywords

Education, pedagogy, microcontrollers, robots.

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